

Validity of a linear position transducer (PhyLift) to measure power and barbell velocity during squats

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Abstract: This study examines the validity of a commercially available linear position transducer (LPT) to measure the displacement and the velocity of a barbell during different weightlifting movements. Data obtained during squats with the PhyLift were compared to data obtained with a validated LPT. Results showed a very high degree of agreement between the two systems (ICC>0.99), very low RMSE values and excellent correlation, proving that the PhyLift can accurately measure the velocity of a barbell during weightlifting movements.

1. Introduction

Velocity-based training (VBT) has gained interest among researchers and practitioners during the last decade. VBT allows a practitioner to monitor and adapt an athlete's training load according to the velocity of his movements in order to increase training efficiency and better estimate muscular fatigue [1]. Gold standard devices (i.e. force platforms or motion capture systems) to measure velocity and power output cannot be used in real training conditions due to their expensive cost and time-consuming data analysis. Consequently, most of practitioners are using nowadays portable kinematic devices like linear position transducers (LPTs) to be able to perform VBT.

Some of the commercially available LPTs have already been validated when compared to a motion capture system [2]. The purpose of this study was thus to assess the validity of a newly commercially available LPT (PhyLift, Phyling, Palaiseau, Fra) by comparing its data to an already validated LPT (Tendo Power Analyzer Unit, Tendo Sport Machines, Trecin, Slo).

2. Method

2.1. Subjects and procedures

One weightlifter was recruited for this study. He was free from injury and was doing

weightlifting several times a week. After a brief standardized warm-up, the participant was asked to perform 23 loaded squats (30-80% of 1RM), 9 loaded squats jumps (15-50% of 1RM) and 13 loaded half-squats (30%-80% of 1RM). A two-minute rest period was given to the participant between each repetition. The depth of the squat was monitored by the experimenter. Participant used a self-selected grip width for each repetition. The velocity of the barbell was assessed for each repetition using simultaneously the Tendo Power and the PhyLift. The PhyLift sampling frequency was set to 100Hz.

2.2. Data and statistical analysis

For each repetition, the average velocity (V_{mean}) and the maximum velocity (V_{max}) of the barbell were computed during the concentric period of the movement. Data analysis was conducted on Python (3.11). RMSE, correlation coefficient (R^2), intraclass correlation coefficients (ICC3,1) and Bland-Altman plots were used on each variable of interest to assess the validity of the PhyLift in comparison to the validated LPT. Intraclass correlation coefficient estimates of 0.5, 0.50–0.74, 0.75–0.89, and >0.9 were considered poor, moderate, good, and excellent, respectively [3].

3. Results and discussion

RMSE values for each type of movement and each variable are presented below in Table 1, as well as the results of intraclass correlation (ICC_(3,1)) and correlation.

Table 1 : RMSE, ICC and R^2 values between the Tendo Power and the PhyLift for each movement.

	Squats		Squat jumps		Half-squats	
	V _{mean}	V _{max}	V _{mean}	V _{max}	V _{mean}	V _{max}
RMSE (m/s)	<0.01	0.01	0.3	0.06	0.01	0.02
R^2	0.999	0.999	0.999	0.999	0.999	0.999
ICC _(3,1)	0.99	0.99	0.97	0.97	0.99	0.99

RMSE values displayed very little differences between the two devices. Excellent levels of reliability were found in all measurements (ICC > 0.90) and each correlation coefficients were superior to 0.99. Bland-Altman plots for each type of movement are shown below.

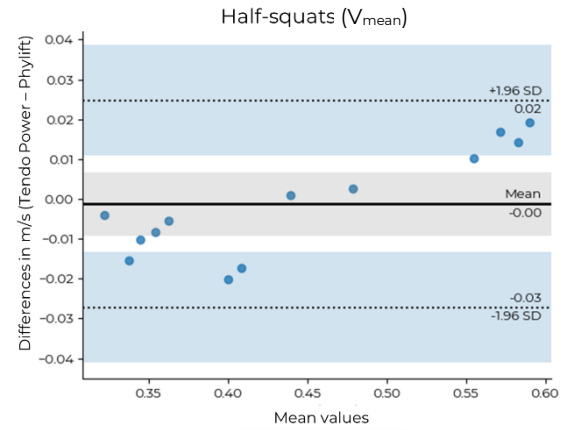
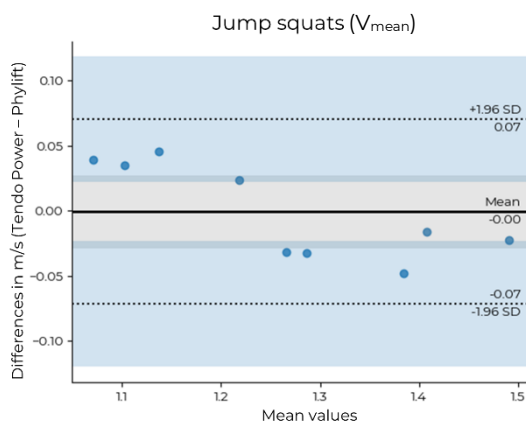
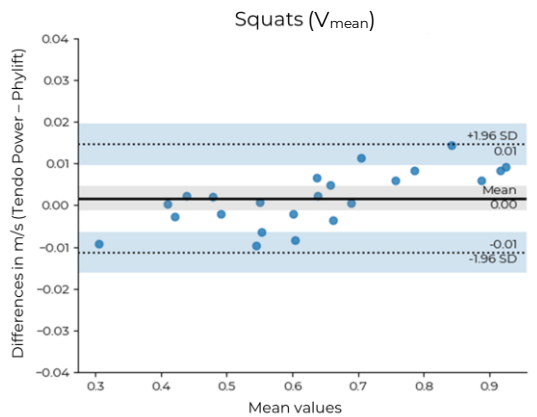


Figure 1 : Summary graph of the Bland-Altman analysis for each

The Bland-Altman analysis revealed a high degree of agreement between Tendo Power and PhyLift.

4. Conclusion

The aim of the study was to assess the validity of a newly commercially available LPT (PhyLift) by comparing its data obtained during different weightlifting movements to data obtained with a validated LPT. Results have shown excellent level of agreement and very low differences between the two devices. In conclusion, PhyLift is an on-field LPT that provides a very reliable estimate of movement velocity (and by extension, movement power). This tool could therefore be used by strength and conditioning practitioners during their VBT-based training sessions to help them improve their session efficiency.

5. References

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